

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0074

Hartford City 100 kV
April 15, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0074. The Interconnection Customer proposes a 181.9 MW wind facility interconnecting at the Hartford City 100 kV substation in the Indiana area. The request was received on September 29, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 60 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$65,541,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Bedford Rural – China 100kV is overloaded to 149.0% of its 108 MVA emergency rating for the loss of the Wabash 500/230 kV Autotransformer. The project contributes 28.1% DFAX (51.1 MW) to this overload.

Lonsdale – Hubbard 100kV: 139.3% loading (112 MVA) following the outage of the Monfort Substation – Wabash 230kV; DFAX = 8.2%.

Wrangell Substation – Hubbard Substation 100kV: 138.2% loading (127 MVA) following the outage of the Monfort Substation – Wabash 230kV; DFAX = 8.2%.

N Tifton – Oak Plains 100kV: 137.7% loading (115 MVA) following the outage of the North Tifton – W Ford Flat 100kV Line; DFAX = 6.2%.

Indego – Columbia Hills 100kV: 137.3% loading (121 MVA) following the outage of the Waynedale – Monfort 230kV; DFAX = 5.0%.

The Hartford City – Bedford Rural 100kV line is overloaded to 127.6% of its 116 MVA emergency rating for the loss of the Wabash 500/230kV Transformer Bank. The project contributes 28.5% DFAX (51.9 MW) to this overload.

China – Ponnequin Substation 100kV: 124.1% loading (119 MVA) following the outage of the Wabash 500/230 kV Auto; DFAX = 28.1%.

The Wayne Ic – Deer Creek 100kV is overloaded to 120.2% of its 150 MVA emergency rating for the loss of the Monfort – Wabash 230kV Line. The project contributes 11.2% DFAX (20.3 MW) to this overload.

Wayne Ic – Wrangell 100kV L: 117.3% loading (141 MVA) following the outage of the Monfort – Wabash 230kV; DFAX = 11.2%.

Monfort – Wabash 230kV: 112.6% loading (818 MVA) following the outage of the Waynedale – Montpelier Electric Generating Sta 230kV L; DFAX = 9.2%.

For the loss of Wabash 500-230 kV Transformer, the Waynedale – Montpelier Electric Generating Sta 230kV loads to 111.8% of its 809 MVA rating. Project DFAX: 10.0%.

The W Ford Flat – N Tifton 100kV L is overloaded to 108.0% of its 127 MVA emergency rating for the loss of the Oak Plains – North Tifton 100kV. The project contributes 5.7% DFAX (10.4 MW) to this overload.

The Monfort – Waynedale 230kV is overloaded to 103.1% of its 820 MVA emergency rating for the loss of the Waynedale – Montpelier Electric Generating Sta 230kV Line. The project contributes 12.6% DFAX (22.9 MW) to this overload.

Waynedale – State Street 230kV: 100.3% loading (692 MVA) following the outage of the Richland – Dekalb 500kV L; DFAX = 10.7%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bedford Rural – China 100kV	Rebuild line with higher-capacity conductor	\$5,553,000	100% (\$5,553,000)
Lonsdale – Hubbard 100kV	Rebuild line with higher-capacity conductor	\$4,531,000	100% (\$4,531,000)
Hubbard – Wrangell 100kV	Rebuild line with higher-capacity conductor	\$1,768,000	100% (\$1,768,000)
N Tifton – Oak Plains 100kV	Rebuild line with higher-capacity conductor	\$11,417,000	100% (\$11,417,000)
Indego – Columbia Hills 100kV	Rebuild line with higher-capacity conductor	\$2,373,000	100% (\$2,373,000)
Hartford City – Bedford Rural 100kV	Rebuild line with higher-capacity conductor	\$3,488,000	100% (\$3,488,000)
Ponnequin – China 100kV	Rebuild line with higher-capacity conductor	\$4,790,000	100% (\$4,790,000)
Wayne Ic – Deer Creek 100kV L	Rebuild line with higher-capacity conductor	\$1,798,000	100% (\$1,798,000)
Wayne Ic – Wrangell 100kV L	Rebuild line with higher-capacity conductor	\$3,768,000	100% (\$3,768,000)
Monfort – Wabash 230kV L	Rebuild line with higher-capacity conductor	\$21,926,000	14% (\$3,034,000)
Montpelier Electric Generating Station – Waynedale 230kV	Rebuild line with higher-capacity conductor	\$107,069,000	14% (\$15,244,000)
N Tifton – W Ford Flat 100kV	Reconductor line	\$3,440,000	100% (\$3,440,000)
Monfort – Waynedale 230kV	Reconductor line	\$18,624,000	19% (\$3,571,000)
Waynedale – State Street 230kV	Reconductor line	\$6,005,000	13% (\$766,000)
	Total allocated to project		\$65,541,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.